DAY 05 OF 16 · 3D & 6DOF STREAMING

One object, many representations

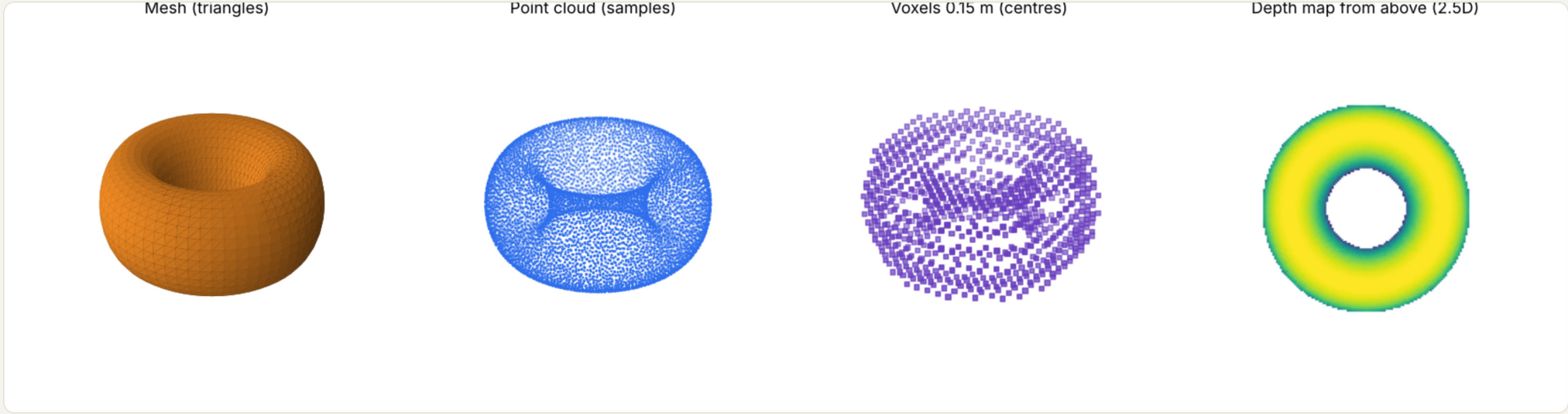
Points, meshes, voxels and depth maps: what each one is good at, and how to convert between them.

Point clouds | Meshes | Voxels | Depth maps | Conversions | Trade-offs

Book: Ch. 4 · Code: day05_representations/

LAB PREVIEW

The same torus in four representations



From representations.py. Each is a different answer to: how do I store a surface?

BOOK CH. 4

Point clouds vs meshes

Point cloud

Free points. Exact samples, fast to transform and render, but no surface between points and high memory.

01

Mesh

Vertices joined by edges into faces (usually triangles). A continuous surface with topology. The standard in games and film.

02

BOOK CH. 4

Voxels and depth maps

Voxels

3D pixels on a grid, like Lego or Minecraft. Store volume, so they suit simulation. Size grows fast as the grid gets finer.

01

Depth map / RGB-D

An image with one distance per pixel. Small and image-friendly, but 2.5D: one value per line of sight.

02

THE CANONICAL LINK

Converting between representations

01

Mesh → points

Sample the surface (uniform or Poisson-disk).

02

Points → voxels

Snap each point to a grid cell; keep occupied cells.

03

Voxels → points

Take each voxel centre: detail below the cell size is lost.

04

Points → depth

Project with a camera; keep the nearest per pixel.

05

Points → mesh

Surface reconstruction, e.g. Poisson (Day 10).

Every arrow loses or invents information. Lab L3 measures how much.

CHOOSING

Which representation for which job?

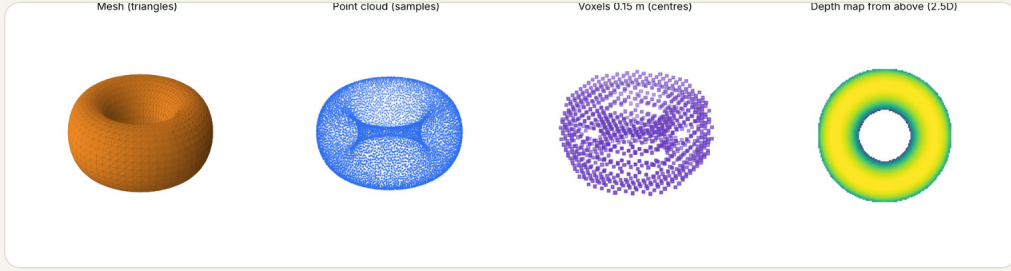
Need	Best fit	Why
Raw capture, measurement	Point cloud	Closest to the sensor, exact samples
Real-time rendering, games, AR	Mesh	GPUs draw triangles; compact for smooth surfaces
Simulation, occupancy, robotics	Voxels	Volume and free space are explicit
Per-frame sensing, video codecs	Depth / RGB-D	Image tools and hardware codecs apply
Photo-real novel views	Gaussian splats (Day 10)	Soft primitives, view-dependent colour

HANDS-ON LAB · COPY, RUN, LOOK

Convert a torus four ways

RUN IT

```
cd day05_representations
python representations.py
```



→ Read the size line for each representation.

→ Rotate the window: points, mesh and voxels side by side.

→ Try voxel sizes 0.5 and 0.02.

Three things to remember

1. Points = samples, mesh = connected surface, voxels = volume grid, depth map = 2.5D image.
2. Converting between them always loses or invents information.
3. Streaming systems pick the representation that is cheapest to send and to draw.

BEFORE NEXT SESSION

Lab L3 questions 1. Next: where does 3D data come from? Bring a phone.